

# Wajid Hussain

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## TECHNICAL SKILLS

- **Programming Languages:** Python, C++, C
- **Databases & Connectors:** MySQL, Microsoft SQL Server, MongoDB, Cassandra, Apache Impala, Kafka Connect
- **Data Warehouses & Data Lakes:** Hive, BigQuery, Databricks, Amazon Redshift
- **Distributed Computing Frameworks:** Apache Hadoop, Apache Spark (PySpark)
- **Messaging & Streaming Platforms:** Confluent Kafka, Google Pub/Sub
- **Workflow Orchestration:** Apache Airflow
- **Visualization & Monitoring:** Power BI, AWS CloudWatch
- **Cloud Platforms (AWS):** S3, EMR, EC2, Lambda, Glue, SNS, SQS
- **Google cloud platform (GCP):** Cloud Pub/Sub, Dataproc, Cloud Storage (GCS), Dataflow, Cloud Composer
- **Microsoft Azure:** Power BI, Databricks, SQL Server
- **Containers & Build Tools:** Docker
- **Version Control & Documentation:** Git

## WORK EXPERIENCE

### Data Analytic Engineer @ Bank Alfalah Limited

July 2025 – Present

#### Key Responsibilities

- **Design, implement, and monitor ETL pipelines** to optimize audit efficiency, data quality, and operational transparency.
- **Extract, clean, and transform data** from SQL databases and Big Data platforms (BAFL's Data Warehouses) in line with defined scope documents.
- **Develop and maintain exception generation processes** using Arbutus Analyzer, SQL, Power BI, and AI/ML tools such as Python.
- **Automate escalation workflows** for branch operations (Conventional/Islamic) to ensure timely issue resolution.
- **Build and enhance in-house exception management systems** to support audit analytics and reporting.
- **Create interactive data visualizations** in Power BI and other BI tools to effectively communicate findings to stakeholders.
- **Deliver Smart Data insights** to on-site audit teams during branch take-ups, enabling real-time decision-making.
- **Develop reusable scripts** in Arbutus Analyzer to reduce repetitive manual tasks and improve audit automation.
- **Provide technical support** and training to audit staff on Arbutus Analyzer and other analytical tools.
- **Explore and implement new features** of Arbutus Analyzer and emerging data engineering technologies to strengthen audit analytics capabilities.
- **Build and deploy ML models** to automate repetitive manual audit tasks, such as anomaly detection, exception classification, and predictive risk scoring.
- **Perform additional tasks** as assigned by the Line Manager, ensuring flexibility and adaptability in dynamic audit environments.

### Data Engineer @ Taj Gasoline Pvt Ltd

Mar 2023 – April 2024

#### Key Responsibilities

- **Design, implement, and monitor ETL pipelines** to extract, transform, and load data from Oracle, Microsoft ERP systems, and Excel files into enterprise data warehouses (Microsoft SQL Server, AWS Redshift).
- **Automated data ingestion and transformation** using Apache Airflow and AWS Glue improving efficiency and reducing manual effort.
- **Managed relational databases** (Microsoft SQL Server, Oracle, MySQL) with a focus on high availability, security, and performance tuning.
- **Developed and optimized SQL queries, stored procedures,** and indexing strategies to improve query execution time across Microsoft SQL Server.
- **Integrated ERP systems** (Oracle ERP & Microsoft Dynamics) into centralized data warehouses, enabling unified business intelligence and analytics.
- **Deployed and managed containerized workflows** with Docker on AWS ECS/EKS, ensuring seamless orchestration and scalability.
- **Administered IT infrastructure including Windows servers,** virtualization, and networking to support enterprise data engineering operations.
- **Implemented and maintained cloud-based solutions in AWS** (S3, EC2, Lambda, CloudWatch, SNS, SQS) to optimizing costs and improving system performance.
- **Ensured data security, backup,** and disaster recovery strategies using AWS Backup and Microsoft SQL Server native tools to guarantee business continuity.
- **Provided technical support for database systems,** cloud platforms, and enterprise applications, ensuring smooth operations across AWS and Microsoft ecosystems.

### Key Responsibilities

- **Provide technical support for enterprise applications**, ensuring high availability, stability, and minimal downtime.
- **Troubleshoot and resolve application, system, and network issues**, to maintain seamless business operations.
- **Monitor and maintain application performance, system logs, and network alerts**, proactively identifying and addressing potential issues.
- **Install, upgrade, and patch enterprise software applications**, ensuring compliance with security and stability standards.
- **Manage user access, authentication, and permissions**, enforcing organizational security policies and regulatory compliance.
- **Maintain and update IT asset inventory**, including hardware, software, and network devices, for accurate resource tracking.
- **Administer and optimize SQL Server and MySQL databases**, ensuring high performance, scalability, and availability.
- **Develop and optimize SQL queries, indexing strategies, and stored procedures** to improve data retrieval efficiency.
- **Implement database backup, recovery, and disaster recovery strategies** to safeguard data and ensure business continuity.
- **Configure and maintain network devices (firewalls, switches, routers)**, ensuring secure and optimal performance.
- **Troubleshoot network connectivity, VPNs, and security protocols**, resolving issues promptly to minimize disruptions.
- **Perform network monitoring, traffic analysis, and performance tuning** to prevent outages and optimize throughput.
- **Ensure data integrity, replication, and failover mechanisms** to support resilient enterprise systems.
- **Collaborate with cross-functional teams to enhance IT infrastructure**, streamline workflows, and support evolving business needs.

### Relevant Projects

#### AAU - Audit Analyzer for Bank Alfalah Audit & Inspection group- Scalable Data Pipeline for Financial Compliance

Description: Architected and implemented a high-performance data pipeline to automate Know Your Customer (KYC) audits across bank branches. The system ingests raw customer datasets containing millions of records (FBR, RTGS, turnover, FTT, EDD, CRP details), applies 107 audit logics, and generates compliance-ready Excel reports in minutes instead of days.

Key Contributions:

- **Data Ingestion & Cleaning:** Automated ingestion of multi-source raw files (Excel/CSV) and applied normalization to handle inconsistent formats and identifiers.
- **Exception Detection:** Implemented 107 audit logics to classify customers into exception categories (e.g., turnover mismatches, occupation vs. purpose contradictions, missing CDD/EDD).
- **Sampling Framework:** Designed dual sampling strategies:
  - Program Sampling → extracts distinct exception accounts for targeted review.
  - Approved Sampling → applies rules to sample new vs. old accounts, further split into high-turnover and other categories.
- **Effectiveness Measurement:** Calculated control effectiveness by analyzing the number of record instances per exception category, providing auditors with quantitative evidence of risk exposure.
- **Performance Optimization:** Reduced audit processing time from 2 days to 5 minutes by leveraging efficient batch transformations in Pandas and modular workflow design.
- **Audit Reporting:** Automated generation of structured Excel reports with exception tables, sampling results, and effectiveness metrics, enabling auditors to complete branch audits rapidly.

Technologies: Python, Pandas, Regex, OpenPyXL, DataFrame transformations, Batch Processing

#### ML Based project for Audit analytic Unit End-to-End Machine Learning Pipeline for Exceptional Case Classification

Description: Developed and deployed a machine learning pipeline to automatically classify branch email responses related to exceptional customer cases. The solution integrated with existing ETL workflows to fetch exceptions from Data servers, route them to branches, and then analyze branch responses to determine resolution status ("Open" vs "Closed").

Key Contributions:

- **Data Ingestion:** Automated extraction of exceptional cases from Big Data servers and routed them to branches via an ETL pipeline.
- **Response Collection:** Integrated Outlook automation to fetch branch email responses tied to unique case identifiers.
- **Data Cleaning & Preprocessing:** Applied regex-based text cleaning to remove signatures, system markers, and noise, while preserving short but meaningful resolution signals.
- **Feature Engineering:** Enriched text with semantic flags (e.g., [RESOLUTION\_CONFIRMED], [PENDING\_ACTION]) to capture resolution and pending action cues.
- **Model Training & Testing:** Fine-tuned transformer models (BERT, Hugging Face) on historical branch responses, then tested on new exceptional case replies to validate accuracy.
- **Classification Pipeline:** Built a daily workflow to classify responses into Open/Closed, merge predictions back into Excel audit reports, and flag unresolved cases for follow-up.
- **Operational Impact:** Reduced manual audit review time, improved transparency in exception handling, and ensured timely escalation of unresolved cases.

Technologies: Python, Pandas, Regex, Hugging Face Transformers (BERT), Scikit-learn, Outlook COM API, Excel (openpyxl), Big Data servers (PySpark/ETL)

### End-to-End Cricket Statistics Pipeline with Google Cloud & Airflow

Description: Designed and deployed a real-time ETL pipeline for cricket statistics using the Google Cloud ecosystem, orchestrated with Apache Airflow (Cloud Composer). The pipeline automated ingestion, transformation, and visualization of live match data, enabling real-time insights for analytics and reporting.

Key Contributions:

- Data Extraction: Built Python scripts to fetch live cricket data via the Cricbuzz API using RESTful calls.
- Workflow Orchestration: Implemented DAGs in Cloud Composer (Airflow) to schedule, monitor, and manage ETL tasks with retries and error handling.
- Cloud Storage: Cleaned and persisted raw data as CSV files in Google Cloud Storage (GCS) for scalable storage.
- Event-Driven Processing: Configured Cloud Functions to trigger downstream jobs automatically upon file uploads.
- Streaming & Transformation: Leveraged Dataflow to process structured data from GCS and load it into BigQuery.
- Visualization: Built an interactive Looker Studio dashboard connected to BigQuery, providing real-time match statistics and insights.

Technologies: Python, Cricbuzz API, Google Cloud Storage (GCS), Cloud Functions, Cloud Composer (Airflow), Dataflow, BigQuery, Looker Studio

### End-to-End ELT Pipeline on Google Cloud – Real-Time Data to Dashboards

Description: Designed and deployed a fully automated ELT pipeline on Google Cloud Platform (GCP) to transform raw operational data into enriched datasets and deliver real-time business insights through interactive dashboards.

Key Contributions:

- Workflow Orchestration: Built DAGs in Cloud Composer (Apache Airflow) to schedule Python scripts for raw data extraction and ingestion into Google Cloud Storage (GCS).
- Data Transformation: Automated pipelines in Cloud Data Fusion to perform data cleaning, PII masking, and application of business logic, ensuring secure and enriched datasets.
- Warehouse Integration: Loaded transformed data into BigQuery, enabling scalable, cost-effective analytics and ad-hoc querying.
- Visualization: Developed interactive dashboards in Looker Studio, providing stakeholders with real-time insights for data-driven decision-making.
- Architecture Design: Delivered a modular, scalable, and secure pipeline built entirely on managed GCP services, reducing operational overhead and ensuring reliability.

Technologies: Python, Cloud Composer (Airflow), Cloud Storage, Cloud Data Fusion, BigQuery, Looker Studio

## Soft Skills

- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking
- Problem-solving
- Public Relations

## EDUCATION

### MS Data Science

Quaid e Awam University of engineering science and technology SBA

May 2024 – Present

### B.E Computer Systems Engineering

Quaid e Awam University of engineering science and technology SBA

Oct 2017 – Dec 2021 Final grade: 3.71 / 4.00 GPA

## CERTIFICATIONS

- AWS Certified Data Engineer – Associate (Udemy)
- Google Cloud Certified Professional Data Engineer (Udemy)
- Google Certified Professional Machine learning Engineer (Udemy)
- Oracle Cloud Infrastructure 2025 Data Science Professional (1Z0-1110-25) (Oracle)
- Data Engineering Fundamentals (Coursera)
- Python Fundamentals for Beginners (Great Learning)
- Networking & Cloud computing (NAVTTTC)

## LANGUAGES

- ENGLISH (FLUENT)
- URDU (FLUENT)